

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 \leq 5$$

$$x_1 \leq 2$$

$$x_1, x_2 \geq 0$$

$$x_1, x_2 \in \mathbb{Z}$$

Rilassamento Lineare

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 \leq 5$$

$$x_1 \leq 2$$

$$x_1, x_2 \geq 0$$

Metodo del simplesso

	x_1	x_2	s_1	s_2	b
s_1	2	2	1	0	5
s_2	1	0	0	1	2
	-2	-1	0	0	0

Entra x_1 ed esce s_2 :

	x_1	x_2	s_1	s_2	b
s_1	2	2	1	0	5
s_2	1	0	0	1	2
	-2	-1	0	0	0

$$R1 \leftarrow R1 - R2 \cdot 2$$

$$R3 \leftarrow R3 + R2 \cdot 2$$

	x_1	x_2	s_1	s_2	b
s_1	0	2	1	-2	1
x_1	1	0	0	1	2
	0	-1	0	2	4

Entra x_2 ed esce s_1 :

	x_1	x_2	s_1	s_2	b
s_1	0	2	1	-2	1
x_1	1	0	0	1	2
	0	-1	0	2	4

$$R1 \leftarrow R1 \cdot \frac{1}{2}$$

$$R3 \leftarrow R3 + R1$$

	x_1	x_2	s_1	s_2	b
x_2	0	1	$\frac{1}{2}$	-1	$\frac{1}{2}$
x_1	1	0	0	1	2
	0	0	$\frac{1}{2}$	1	$\frac{9}{2}$

Forma Standard

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 + s_1 = 5$$

$$x_1 + s_2 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0$$

$$x_1 = 2$$

$$x_2 = \frac{1}{2}$$

$$s_1 = 0$$

$$s_2 = 0$$

La soluzione $x_{RL} = (2, \frac{1}{2})$ non è ammissibile per un problema di PLI.

Taglio di Gomory frazionario

Si crea il nuovo vincolo dato dal taglio:

$$x_B + \sum_j (a_j - \lfloor a_j \rfloor) x_j = b - \lfloor b \rfloor$$

$$\{a\} = a - \lfloor a \rfloor$$

$$\sum_j \{a_j\} x_j \geq \{b\}$$

$$x_2 + \frac{1}{2} s_1 - s_2 \geq \frac{1}{2}$$

$$(1 - \lfloor 1 \rfloor) x_2 + \left(\frac{1}{2} - \left\lfloor \frac{1}{2} \right\rfloor \right) s_1 - (1 - \lfloor 1 \rfloor) s_2 \geq \frac{1}{2} - \left\lfloor \frac{1}{2} \right\rfloor$$

$$\frac{1}{2} s_1 \geq \frac{1}{2}$$

$$s_1 \geq 1$$

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 + s_1 = 5$$

$$x_1 + s_2 = 2$$

$$s_1 \geq 1$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Metodo del Simplex Duale

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 + s_1 = 5$$

$$x_1 + s_2 = 2$$

$$-s_1 + s_3 = -1$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

	x_1	x_2	s_1	s_2	s_3	b
x_2	0	1	$\frac{1}{2}$	-1	0	$\frac{1}{2}$
x_1	1	0	0	1	0	2
s_3	0	0	-1	0	1	-1
	0	0	$\frac{1}{2}$	1	0	$\frac{9}{2}$

Entra s_1 ed esce s_3 :

Forma Standard

$$\min f(x) = -2x_1 - x_2$$

$$2x_1 + 2x_2 + s_1 = 5$$

$$x_1 + s_2 = 2$$

$$s_1 - s_3 = 1$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

	x_1	x_2	s_1	s_2	s_3	b
x_2	0	1	$\frac{1}{2}$	-1	0	$\frac{1}{2}$
x_1	1	0	0	1	0	2
s_3	0	0	-1	0	1	-1
	0	0	$\frac{1}{2}$	1	0	$\frac{9}{2}$

$$\begin{aligned} \text{R1} &\leftarrow \text{R1} + \text{R3} \cdot \frac{1}{2} \\ \text{R4} &\leftarrow \text{R4} + \text{R3} \cdot \frac{1}{2} \\ \text{R3} &\leftarrow \text{R3} \cdot -1 \end{aligned}$$

	x_1	x_2	s_1	s_2	s_3	b
x_2	0	1	0	-1	$\frac{1}{2}$	0
x_1	1	0	0	1	0	2
s_1	0	0	1	0	-1	1
	0	0	0	1	$\frac{1}{2}$	4

$$\begin{aligned} x_1 &= 2 \\ x_2 &= 0 \\ s_1 &= 1 \\ s_2 &= 0 \\ s_3 &= 0 \end{aligned}$$

$$\min f(x) = -2x_1 - x_2 \rightarrow -2(2) - (0) = -4 - 0 = \boxed{-4}$$